Section: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3460**

Title: **SHEAR WEBS**

Revision: **0.1**

Date: **03/03/2020**

1 INTRODUCTION

1.1 METHOD-SUMMARY

This chapter describes the methodology to manage rectangular flat or curved composite panels after buckling under shear loads. The proposed method extends the applicability of the diagonal tension theory developed for isotropic materials.

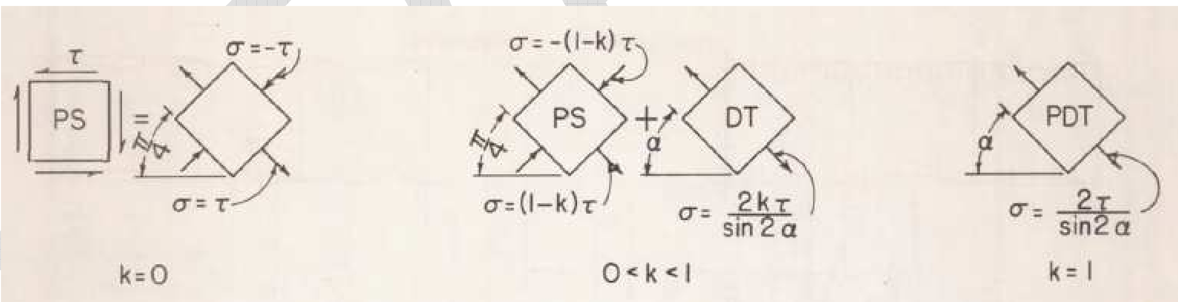


Figure 1. Shear resistant and diagonal tension webs, from Kuhn's report (Ref. 1)

1.2 SCOPE AND LIMITATIONS

- Stress analysis of composite shear webs and panels
- Rectangular flat panels of curved shells (one direction of curvature)
- Panels with holes of significant diameter not covered (however, pilot or rivet holes can be treated locally after the analysis in this chapter is performed)

2 METHOD DESCRIPTION

2.1 BUCKLING LOADS

The onset of buckling shall be calculated for the combined loads case (not only considering pure shear) according to methods described in chapter §3450 “PLATE/SHELL BUCKLING”. The buckling level is defined as:

BL = buckling_load / applied_load

2.2 POST-BUCKLING IN SHEAR: DIAGONAL TENSION

In first approximation, the available theory for plane panels made of isotropic materials shall be used (see Ref. 1). Following discussion is make assuming $a > b$ (in the opposite case replace in expressions *a* by *b* and *x* by *y* , and vice versa).

The diagonal tension factor is obtained as follows (Kuhn, Ref. 1, eq. (50)):

$k = \tanh \{ [0.5 + (300 \cdot t \cdot a)/(R \cdot b)] \cdot \log_{10}(1/BL) \}$ if $BL < 1.0$, otherwise $k = 0.0$

… where *BL* is the Buckling Level, i.e. the ratio obtained by dividing the buckling load by the applied load (considering the combined load system with all its components). The inverse of the *BL* ratio (*1/BL*) corresponds, in pure shear loads, to the τ/τ_{cr} in Kuhn’s report.

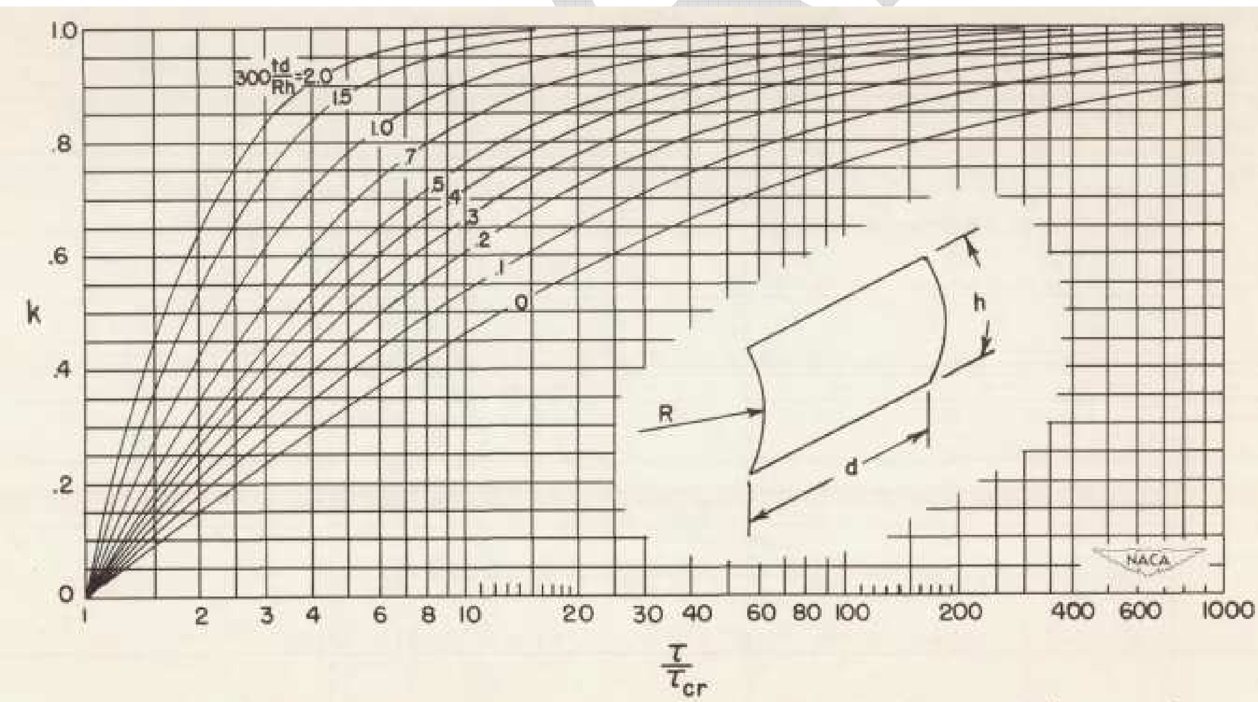


Figure 2. Diagonal factor in curved panels (Kuhn’s report TN 2661, figure 13)

NASA investigation on composite panels in diagonal tension (see Ref. 2) suggests simply using:

$k = 1 - BL$

The diagonal tension angle α will be assumed equal to 45º, a reasonably good approximation according to references consulted (the process to get its value consistently with theory can be found in literature).

Forced compression loads on stiffeners

The forced compression on each longitudinal stiffener, induced by the diagonal tension at the adjacent subpanel subjected to shear N_{xy} , per Kuhn's eq. (51) (adding effective skin to the stringer section) will be expressed as:

$$\Delta P_x = -k \cdot N_{xy} \cdot b \cdot \cot(\alpha) = -k \cdot N_{xy} \cdot b / \tan(\alpha) \approx -k \cdot N_{xy} \cdot b$$

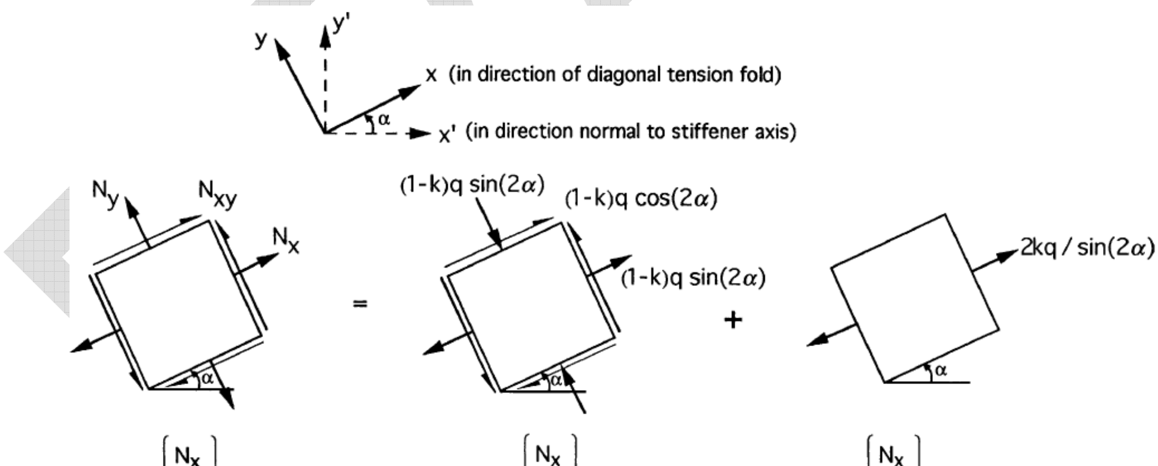
Similarly, on transversal stiffeners (Kuhn's eq. (52)):

$$\Delta P_y = -k \cdot N_{xy} \cdot a \cdot \tan(\alpha) \approx -k \cdot N_{xy} \cdot a$$

These compression forces induced by diagonal tension shall be added to primary loads in the stiffeners limiting the panel.

Final loads in skin

The initial shear load in the panel q , after the calculation of the buckling onset, is treated separately to perform the (incomplete) diagonal tension analysis described in literature and summarized in previous section. From the diagonal tension angle α and factor k , the post-buckled loads, expressed in a reference system parallel to the diagonal tension direction can be calculated…



$$\begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix} = \begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix}_{SR} + \begin{Bmatrix} N_x \\ N_y \\ N_{xy} \end{Bmatrix}_{DT} = \begin{Bmatrix} (1-k)q\sin(2\alpha) \\ -(1-k)q\sin(2\alpha) \\ (1-k)q\cos(2\alpha) \end{Bmatrix}_{SR} + \begin{Bmatrix} 2kq/\sin(2\alpha) \\ 0 \\ 0 \end{Bmatrix}_{DT}$$

Figure 3. Final redistribution of load in skin in incomplete diagonal tension (Ref. 2)(*)

(*) Note the notation changes in figure:

- Original applied shear load N_{xy} is denoted as q ($q = \tau \cdot t$) in figure
- Additionally, N_k' symbols in figure denote plate loads in the global reference axis system (axis parallel to panel sides) and N_k denote load in a local reference axis system with X parallel to the diagonal tension direction

Post-buckling loads in incomplete diagonal tension shall be then expressed in the global reference axis system and added to X and Y axial loads segregated after buckling analysis.

Strength check shall be done then with the full set of loads in post-buckled panel by using the approved failure theories.

In NASA report there are also discussions on…

- Skin bending at toe
- Peel stress analysis

Add that ? Look also in Kassapoglou's book

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	*Units*	*Definition*
a	mm	Length of panel (in X / axial direction)
α	º	Diagonal tension angle
b	mm	Width of panel (in Y / arc direction)
cr	-	As subscript, refers to critical (typically to buckling onset level)
k	-	Diagonal tension factor
N_x, N_y, N_{xy}	N/mm	In-plane loads in the panel expressed in the global reference axis system of axis parallel to the panel sides (note that in figures shown from external sources the reference axis are rotated)
t	mm	Thickness
q	N/mm	In-plane shear load in panel
R	mm	Radius of curvature (around X axis)
TBC		To Be Confirmed
τ	MPa	Shear stress (equal to q/t)
Z	-	Curvature parameter

REFERENCES

	Document Reference	*Issue*	*Title*
Ref. 1	[P. Kuhn et al, NADA] TN-2661	1952	A SUMMARY OF DIAGONAL TENSION. PART I – METHODS OF ANALYSIS.
Ref. 2	[NASA] CR-97-20656	1997	Novel composites for wing and fuselage applications – speedy nonlinear analysis of postbuckled panels in shear (SNAPPS)
Ref. 3	[Christos Kassapoglou]	2nd Ed.	Design and Analysis of Composite Structures

SOFTWARE

Program	*Version*	*Description*

Table 1. *Software programs referenced in this chapter*

RECORD OF REVISIONS

Revision Date	*Authors*	*Change Reason*	*Pages/Sub-chapters affected*
0.1 03/03/2020	G. Baños	First issue	All pages

Table 2. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

Dpt. \ Site	*Getafe*	*Manching*
Stress Methods (TEASP-TL3)	G. Baños 03/03/2020	F. Glock DD/MM/2020

Table 3. *Approval signatures table for last revision of this chapter*